

RCS-VRF Subnet: Mathematical Reference

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Abstract

Reference document capturing the mathematical content underpinning the RCS-VRF Subnet protocol. Covers F_{XEB} statistics, Haar-randomness and Porter–Thomas, min-entropy and the Leftover Hash Lemma, the Aaronson–Hung 2023 bound (and its Liu et al. 2025 correction), the heavy-output attack family, the VRF construction, and the hash-chained audit log. Not a full derivation of every theorem — focused on the relationships and formulas the protocol depends on, with derivations included where they are short or illuminating.

Notation.

- n = number of qubits
- $N = 2^n$ = dimension of the Hilbert space
- m = number of samples (shots)
- M = total number of circuits in a multi-circuit protocol
- $p_C(x) = |\langle x|C|0^n\rangle|^2$ = ideal probability of measuring bitstring x from circuit C
- F = fidelity of the quantum sampling
- ε = soundness parameter
- H_∞ = min-entropy; H = Shannon (von Neumann) entropy

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1 The F_{XEB} statistic

1.1 Definition

Given m samples $\{x_i\}_{i=1}^m$ from a quantum device running circuit C , the Linear Cross-Entropy Benchmarking score (LXEB or F_{XEB}) is:

$$F_{\text{XEB}} = \frac{2^n}{m} \sum_{i=1}^m p_C(x_i) - 1 = N \cdot \langle p_C(x_i) \rangle_i - 1 \quad (1)$$

The ideal probabilities $p_C(x_i)$ are computed classically (full state-vector simulation up to $n \approx 25$, tensor-network contraction beyond).

1.2 Expected values under different sampling strategies

Uniform sampling. If samples are drawn uniformly from $\{0, 1\}^n$:

$$\mathbb{E}[\langle p(x_i) \rangle] = \mathbb{E}_x[p(x)] = \frac{1}{N} \quad (2)$$

$$\mathbb{E}[F_{\text{XEB}}] = N \cdot \frac{1}{N} - 1 = 0 \quad (3)$$

Ideal quantum sampling. Samples drawn from p_C . The expected value of $\langle p(x_i) \rangle$ is the *collision probability* of p_C :

$$\mathbb{E}[\langle p(x_i) \rangle] = \sum_x p(x)^2 \quad (4)$$

For a Haar-random circuit (see §2):

$$\mathbb{E}_{\text{Haar}} \left[\sum_x p(x)^2 \right] = \frac{2}{N+1} \approx \frac{2}{N} \quad (5)$$

$$\mathbb{E}[F_{\text{XEB}}] = N \cdot \frac{2}{N} - 1 = 1 \quad (6)$$

Quantum sampling at fidelity $F < 1$. Model the noisy state as a depolarising mixture:

$$\rho_F = F \cdot |\psi\rangle\langle\psi| + (1-F) \cdot \frac{I}{N} \quad (7)$$

The expected $p(x_i)$ under this mixture is:

$$\mathbb{E}[p(x_i)] = F \cdot \frac{2}{N} + (1-F) \cdot \frac{1}{N} = \frac{1+F}{N} \quad (8)$$

$$F_{\text{XEB}} = N \cdot \frac{1+F}{N} - 1 = F \quad (9)$$

Linear fidelity relation. $F_{\text{XEB}} \approx F$. This is the defining property that makes F_{XEB} a useful fidelity *measurement*, not just a yes/no test.

1.3 Variance and statistical convergence

For m samples, the standard deviation of F_{XEB} scales as $\sim 1/\sqrt{m}$. At $m = 1000$ (our v0.x parameter), the shot-noise contribution to $\sigma(F_{\text{XEB}})$ is $\sim 3\%$ of the mean.

Per-circuit variance (across different random circuits with same n, d) is typically larger than shot noise. Our depth sweep at $n = 8, d = 10$ measured noiseless $F_{\text{XEB}} \approx 1.0 \pm 0.2$ across multiple circuits — circuit-to-circuit variance dominates shot noise at this scale.

2 Haar-randomness and Porter–Thomas

2.1 Haar measure

The *Haar measure* is the unique probability distribution on the unitary group $U(N)$ that is invariant under multiplication by any fixed unitary — the quantum analogue of “uniform random” on the unitary group.

A Haar-random circuit is one whose unitary is sampled from the Haar measure. Properties computed under the Haar measure are reference values the protocol calibrates against.

2.2 Porter–Thomas distribution

For a Haar-random circuit acting on $|0^n\rangle$, the output probabilities $p(x)$ follow the Porter–Thomas distribution:

$$f(p) = N \cdot e^{-N \cdot p}, \quad \Pr[p(x) \leq t] = 1 - e^{-Nt} \quad (10)$$

This is an exponential distribution with mean $1/N$. Key moments:

$$\mathbb{E}[p(x)] = \frac{1}{N} \quad (\text{mean — same as uniform}) \quad (11)$$

$$\text{Var}[p(x)] = \frac{1}{N^2} \quad (\text{variance — much larger than uniform's}) \quad (12)$$

$$\mathbb{E}\left[\sum_x p(x)^2\right] = \frac{2}{N+1} \quad (\text{collision probability — twice uniform's } 1/N) \quad (13)$$

The factor of 2 in the collision probability is what gives $F_{\text{XEB}} \approx 1$ for honest noiseless quantum sampling.

2.3 Why Porter–Thomas matters for the protocol

PT is not a passive consequence of using random circuits — every quantitative claim in the protocol is calibrated to PT statistics:

Protocol mechanism	What it needs from PT
$F_{\text{XEB}} \approx 1$ for honest	$\mathbb{E}_{\text{PT}}[\sum p^2] = 2/N$
$F_{\text{XEB}} = F$ for noisy	Linear mixture of PT and uniform
Quantum-classical hardness gap	PT's heavy outcomes are classically unpredictable
Min-entropy $\approx n - \log n$	PT's specific H_∞ structure
Heavy-output attack analysis	Defined relative to PT's heavy outcomes

If the circuit distribution deviates from PT (e.g., undertrained circuit family, shallow depth), all these calibrations break. Empirically, our v0 depth sweep found noiseless $F_{\text{XEB}} \approx 3\text{--}7$ at $d = 4$ — direct evidence the $d = 4$ circuit family was not producing PT statistics.

2.4 Approximate t -designs

A truly Haar-random unitary on n qubits requires exponentially many parameters to specify — no finite gate set can realise it exactly. RCS approximates Haar using finite circuit families. The relevant notion is *approximate t -design*:

Definition 2.1. A circuit family is an ε -approximate t -design if its first t moments match Haar's first t moments to error ε .

For F_{XEB} (which depends on second moments — collision probabilities), we need approximately a *2-design*. The AH 2023 / Liu et al. 2025 analysis nominally requires sufficient depth that the circuit family is close to a 2-design.

2.5 Empirical convergence

The depth at which a circuit family becomes a good 2-design depends on connectivity:

- *1D brickwork (nearest-neighbor)*: depth $\sim n$ needed for 2-design
- *All-to-all random pairing (Liu et al. recipe)*: depth $\sim \log n$ needed for 2-design

Our v0 depth sweep at $n = 8$ found:

d	Noiseless F_{XEB}	Noisy F_{XEB}	Interpretation
4	3.1	0.7	Far from Haar
8	1.06 ± 0.09	0.38 ± 0.06	Close to Haar
10	1.11 ± 0.09	0.29 ± 0.03	Close to Haar (Liu et al. choice)
14	0.96 ± 0.05	0.12 ± 0.02	Close to Haar (high noise)

We use $d = 10$ with all-to-all connectivity as v0.x's default — confirmed empirically to put the circuit family close enough to Haar for PT statistics to hold.

3 F_{XEB} CDFs under realistic sampling

3.1 Uniform samples

For m samples drawn uniformly from $\{0, 1\}^n$, the sum $\sum p(x_i)$ follows an Erlang distribution of shape m :

$$\Pr[F_{\text{XEB}} \leq \chi \mid \text{uniform}] = \tilde{\Gamma}(m, m(\chi + 1)) \quad (14)$$

where $\tilde{\Gamma}$ is the regularised lower-incomplete Gamma function.

3.2 Ideal quantum samples

For m samples drawn from the ideal PT distribution, $\sum p(x_i)$ follows an Erlang distribution of shape $2m$:

$$\Pr[F_{\text{XEB}} \leq \chi \mid \text{quantum}] = \tilde{\Gamma}(2m, m(\chi + 1)) \quad (15)$$

3.3 Mixed-fidelity samples

A fidelity- F sample is modelled as drawn from the PT distribution with probability F and uniform with probability $1 - F$. The CDF over m mixed samples is (Liu et al. 2025 Eq. III.12):

$$\Pr[F_{\text{XEB}} \leq \chi \mid \text{fidelity } F] = \sum_{\ell=0}^m \tilde{\Gamma}(m + \ell, m(\chi + 1)) \binom{m}{\ell} F^\ell (1 - F)^{m-\ell} \quad (16)$$

This is the formula used to compute $Q_{\min}$ in the security analysis (§5).

4 Min-entropy and the Leftover Hash Lemma

4.1 Min-entropy definition

For a discrete distribution P over X :

$$H_\infty(X) = -\log_2 \left(\max_x P(X = x) \right) \quad (17)$$

This is the worst-case unpredictability. If $H_\infty = k$, no specific outcome has probability greater than 2^{-k} .

Remark. Operational meaning. An attacker's best guessing strategy has success probability 2^{-H_∞} . Min-entropy directly bounds guessability, which is exactly what cryptographic applications need.

4.2 Min-entropy vs. Shannon entropy

For a uniform distribution, $H_\infty = H = \log_2 N$. For non-uniform distributions, $H_\infty \leq H$, with the gap potentially large.

Example. Consider $P(00\dots 0) = 1/2$ and $P(\text{other}) = 0.5/(N-1)$.

$$H \approx 1 + 0.5 \log_2(N-1) \approx n/2 \quad (\text{looks decent}) \quad (18)$$

$$H_\infty = -\log_2(0.5) = 1 \quad (\text{terrible}) \quad (19)$$

The Leftover Hash Lemma requires *min-entropy*, not Shannon. If we used Shannon, extraction on this distribution would collapse half the time — the opposite of uniform.

4.3 Min-entropy for Haar / Porter–Thomas

For a single sample drawn from a Haar-random circuit:

$$H_\infty(\text{PT}) \approx n - \log_2(n) - \log_2(\ln 2) \approx n - \log_2(n) - 0.529 \quad (20)$$

The derivation uses the fact that the maximum probability under PT is approximately $n \cdot \ln(2)/2^n$.

For $n = 8$: $H_\infty^{\text{Haar}} \approx 8 - 3 - 0.53 \approx 5.5$ bits per sample.

4.4 The Leftover Hash Lemma

Theorem 4.1 (Leftover Hash Lemma). *Let X be a random variable with $H_\infty(X) \geq k$. Let $\mathcal{H}$ be a 2-universal hash family from X to $\{0,1\}^\ell$. For uniformly random $h \leftarrow \mathcal{H}$, the joint distribution of $(h, h(X))$ is ε -close to uniform on $\mathcal{H} \times \{0,1\}^\ell$, where:*

$$\varepsilon = \frac{1}{2} \sqrt{\frac{2^\ell}{2^k}} \quad (21)$$

Solving for ℓ in terms of ε :

$$\ell \leq k - 2 \log_2\left(\frac{1}{\varepsilon}\right) \quad (22)$$

This is the maximum number of bits we can safely extract.

4.5 Toeplitz extractor as a 2-universal hash

The Toeplitz extractor maps n input bits to ℓ output bits via $T \cdot x \bmod 2$, where T is an $\ell \times n$ Toeplitz matrix (constant along anti-diagonals). The matrix is generated from a fresh random seed of length $\ell + n - 1$ bits.

The Toeplitz family is 2-universal: for any distinct $x \neq y$:

$$\Pr_T[T \cdot x = T \cdot y] \leq 2^{-\ell} \quad (23)$$

which is exactly the condition the LHL needs.

In our protocol:

- Input: 8000 bits (1000 samples $\times$ 8 qubits) from quantum measurement
- Seed: from drand pulse, expanded via SHAKE256 to needed length
- Output: 64 bits (in v0.x; production target depends on H_∞ bound)

The drand pulse serves as the LHL’s “uniformly random hash selection.” Crucially, drand is independent of the quantum samples, which is what the LHL requires.

5 The Aaronson–Hung bound

5.1 What the bound aims to prove

The AH 2023 protocol’s goal: given that samples pass F_{XEB} at score $\geq \chi$, lower-bound the min-entropy of those samples (conditional on the circuit).

If proved, this lets us:

1. Run XEB to certify the samples are “good enough”
2. Compute a min-entropy bound from the F_{XEB} score
3. Apply the LHL to extract uniform bits

Without such a bound, passing F_{XEB} does not translate into a quantitative randomness claim.

5.2 The corrected single-round bound (von Neumann entropy)

AH 2023’s original Theorem 5.8 claimed to lower-bound *min-entropy* directly. Liu et al. 2025 identified a flaw: the theorem as stated is impossible.

Remark (Counterexample to original AH min-entropy claim). An algorithm that samples honestly 99% of the time and returns the all-zeros bitstring 1% of the time has $H_\infty = -\log_2(0.01) \approx 6.6$ bits regardless of n , and passes LXEB with high probability.

Liu et al.’s fix: bound *von Neumann entropy* in the single-round analysis, then derive smooth min-entropy at the protocol level via the Entropy Accumulation Theorem.

Theorem 5.1 (Liu et al. 2025 Theorem 6, corrected single-round bound). *Under the LLHA hardness assumption, for any device A solving $\text{MLXEB}_{b,k}(\mathcal{D})$ with probability q :*

$$H(Z \mid \vec{C}) \geq \frac{B}{2} \left(\frac{bq - 1}{b - 1} \right) - n^{-\omega(1)} \quad (24)$$

where Z = device output (k bitstrings); $\vec{C}$ = input circuits; B = LLHA parameter ($B \leq n/2$ by Grover); b = LXEB threshold (corresponds to $1 + F_{\text{XEB}}$); q = probability of passing LXEB.

This bounds Shannon / von Neumann entropy, not min-entropy directly.

5.3 The protocol-level smooth min-entropy bound

Theorem 5.2 (Liu et al. 2025 Theorem 1). *Let Ω denote the event that the protocol does not abort. Given $\varepsilon_s \in (0, 1/4)$, the protocol either aborts with probability $> 1 - 4\varepsilon_s$, or:*

$$\boxed{H_{\min}^{\varepsilon_s}(X_M \mid \tilde{I}_{\text{sn}}) \geq Q_{\min}(n - 1) + \log \varepsilon_s} \quad (25)$$

where X_M = the M bitstrings collected; $\tilde{I}_{\text{sn}}$ = classical side information; $Q_{\min}$ = minimum quantum rounds the adversary must execute for the protocol to pass with probability $\geq 4\varepsilon_s$.

The per-quantum-round smooth min-entropy contribution is $n - 1$ bits, plus a $\log \varepsilon_s$ “soundness tax.”

5.4 Computing $Q_{\min}$

$Q_{\min}$ is the result of an optimisation over adversary strategies. The adversary’s optimal strategy uses Q quantum samples (honest) and $(M - Q)$ classical samples simulated at effective fidelity:

$$\phi_{\text{classical}} = \frac{A \cdot t_{\text{threshold}}}{B} \quad (26)$$

where A is adversary FLOPS, B is per-circuit simulation cost, $t_{\text{threshold}}$ is the protocol's time threshold.

Using the mixed-fidelity CDF (16), the adversary's success probability is:

$$\varepsilon_{\text{adv}}(Q, \chi) = 1 - \sum_{\ell=0}^m \tilde{\Gamma}(m + \ell, m(\chi + 1)) \binom{m}{\ell} \phi_{\text{eff}}^{\ell} (1 - \phi_{\text{eff}})^{m-\ell} \quad (27)$$

where $\phi_{\text{eff}}(Q) = \frac{Q}{M} + \frac{M-Q}{M} \phi_{\text{classical}}$. Then:

$$Q_{\min} = \arg \min_Q \{Q : \varepsilon_{\text{adv}}(Q, \chi) \geq 4\varepsilon_s\} \quad (28)$$

5.5 Honest-server simplification

For a server that does all M rounds honestly (no classical mixing), $Q_{\min} = M$ and (25) becomes:

$$H_{\min}^{\varepsilon_s} \geq M(n - 1) + \log \varepsilon_s \quad (29)$$

For our v0.3a parameters ($n = 8$, $M = 1000$, $\varepsilon_s \approx 2^{-34}$):

$$H_{\min}^{\varepsilon_s} \geq 1000 \cdot 7 + \log_2(2^{-34}) \approx 7000 - 34 = 6966 \text{ bits} \quad (30)$$

After LHL with $\varepsilon = 2^{-32}$:

$$\text{max extractable} \approx 6966 - 64 = 6902 \text{ bits} \quad (31)$$

Remark (Caveat at v0.3a scale). At $n = 8$, classical simulation is trivial (microseconds on a laptop). For any reasonable adversary classical power A , $\phi_{\text{classical}} \approx 1$, forcing $Q_{\min} \approx 0$ and the bound to ≈ 0 . The honest-server simplification only makes sense as a “what would the bound say if we trusted the server” exercise; real security at $n = 8$ is not achievable. The protocol's security activates at larger n where classical simulation costs become prohibitive (Liu et al. 2025 demonstrates this experimentally at $n = 56$).

5.6 Comparison with v0.3a's heuristic

v0.3a uses a simplified bound:

$$H_{\infty} \approx m \cdot F_{\text{XEB}} \cdot (n - \log_2 n - \log_2 \ln 2) \quad (32)$$

At our parameters: ≈ 1660 bits. The Liu et al. honest-server bound: ≈ 6966 bits.

Bound	H_{∞} (bits)	Max extractable (bits)	Ratio
v0.3a heuristic $m \cdot F(n - \log n - \log \ln 2)$	1660	1596	1.0×
AH/Liu corrected (honest server) $M(n - 1) + \log \varepsilon_s$	6966	6902	4.3×

Differences:

- *Magnitude*: v0.3a is $\sim 4\times$ conservative.
- *Functional form*: v0.3a scales with F_{XEB} ; the real bound has F_{XEB} entering only through $Q_{\min}$, not as a per-sample multiplier.
- *Per-quantum-sample entropy*: v0.3a uses $(n - \log n - \log \ln 2)$; the real bound uses $(n - 1)$.
v0.3a's bound is *safe* (conservative) but has the wrong functional form. v0.3a-revised should adopt the honest-server bound or full $Q_{\min}$ computation.

6 The heavy-output attack

6.1 Why PT enables this attack

Porter–Thomas has a tail of “heavy” bitstrings — outcomes with probability much higher than $1/N$. An attacker who can identify heavy outcomes (e.g., by classical simulation) can submit only those, achieving F_{XEB} much higher than honest quantum sampling.

For the top- k strategy: attacker sorts bitstrings by $p(x)$, submits the top k . Expected F_{XEB} :

$$F_{\text{XEB}}^{\text{top-}k} = \frac{N}{k} \sum_{x \in \text{top-}k} p(x) - 1 \quad (33)$$

For PT, the top fraction k/N captures a much larger probability mass than k/N would suggest (heavy-tail effect), making $F_{\text{XEB}} \gg 1$.

6.2 Empirical values from v0 spoofing experiment

At $n = 8$, $d = 10$, 5 trials per strategy:

Attack strategy	Mean F_{XEB}	Std
Honest noiseless	1.08	0.23
Honest noisy ($F = 0.5$)	0.30	0.08
Uniform random (circuit-blind)	0.005	0.009
Sample from p (perfect classical)	1.07	0.22
Top-1 guess	5.31	1.43
Top-4 guess	4.37	0.99
Top-16 guess	3.18	0.50

6.3 v0.1’s two-sided defence

v0.1 added an upper bound: $\chi_{\text{low}} \leq F_{\text{XEB}} \leq \chi_{\text{high}}$. At our parameters: $\chi_{\text{low}} = 0.15$, $\chi_{\text{high}} = 2.5$.

Strategy	v0 (one-sided) pass rate	v0.1 (two-sided) pass rate
Honest noiseless	100%	100%
Honest noisy	100%	100%
Uniform random	0%	0%
Top-1 guess	100%	0%
Top-4 guess	100%	0%
Top-16 guess	100%	20%

The upper bound closes the heavy-output attack vector for top-1 and top-4 perfectly, and substantially for top-16.

6.4 The remaining attack: perfect classical sampling

The “sample from p ” strategy gives $F_{\text{XEB}} \approx 1.07$ — indistinguishable from honest noiseless quantum. This attack:

- Requires $O(2^n)$ compute to enumerate all amplitudes
- Is trivially feasible at $n = 8$ ($N = 256$)
- Is infeasible at $n = 56$ ($N \approx 7 \times 10^{16}$, Frontier-scale)

The defence is *parameter scale*, not the F_{XEB} bounds. This is why the LLHA hardness assumption is required: no efficient classical algorithm for this task at appropriate parameter scales.

7 The VRF construction

7.1 v0.2 / v0.3a: Sign + Hash

For each participant i with Ed25519 keypair (sk_i, pk_i) :

$$\pi_i = \text{Ed25519_sign}(sk_i, \alpha) \quad (\text{proof}) \quad (34)$$

$$\beta_i = \text{SHA3-256}(\pi_i) \quad (\text{VRF output}) \quad (35)$$

where α is the round's public input.

Verification. Given $(pk_i, \alpha, \beta_i, \pi_i)$:

1. Check $\text{Ed25519_verify}(pk_i, \alpha, \pi_i)$ succeeds.
2. Check $\text{SHA3-256}(\pi_i) = \beta_i$.

Properties.

- *Deterministic*: same (sk_i, α) always gives same (π_i, β_i) (Ed25519 per RFC 8032 is deterministic).
- *Verifiable*: any party with pk_i can check both conditions.
- *Unpredictable without sk_i* (under the assumption SHA3-256 is a good PRF over Ed25519 signatures).

What this construction is NOT.

- Not full RFC 9381 ECVRF, which has a more elaborate proof structure giving stronger formal pseudorandomness guarantees.
- Not formally proved pseudorandom — depends on SHA3-256 destroying the algebraic structure of Ed25519 signatures.

7.2 H + SHAKE256 aggregation

After collecting n_p VRF outputs $(\beta_1, \dots, \beta_{n_p})$:

$$c = \beta_1 \parallel \beta_2 \parallel \dots \parallel \beta_{n_p} \quad (\text{concatenation}) \quad (36)$$

$$\text{seed} = \text{SHA3-256}(c) \quad (H \text{ step} — \text{compress to 32 bytes}) \quad (37)$$

$$\text{bytes} = \text{SHAKE256}(\text{seed}) \quad (\text{XOF} — \text{expand to arbitrary length}) \quad (38)$$

The 32-byte seed is what LEG 3 uses as input to circuit generation. SHAKE256 expands it into all the bytes needed for random angles, gate selections, and pairings.

The compress-then-expand pattern provides a clean cryptographic boundary independent of how many participants there are or what their contributions look like (collusion-resistance rationale).

8 Hash chains and the audit log

8.1 v0.x audit log structure

Each entry has the form:

$$\text{entry}_i = (\text{leg}_i, \text{data}_i, h_{i-1}, h_i, t_i) \quad (39)$$

where:

$$h_i = \text{SHA3-256}(\text{serialize}(\text{leg}_i, \text{data}_i, h_{i-1})) \quad (40)$$

$$h_0 = 0^{256} \quad (\text{chain root}) \quad (41)$$

8.2 What single-chain hashing detects

For an attacker who modifies one entry’s data field:

- The entry’s recomputed hash does not match the stored h .
- The next entry’s `prev_hash` no longer matches.
- Multiple consistency checks fail.

This catches *casual tampering* — single-point modifications without re-doing the chain math.

8.3 What single-chain hashing does NOT detect

For a sophisticated attacker willing to redo work:

1. Modify entry j ’s data.
2. Recompute h_j .
3. Update entry $j + 1$ ’s `prev_hash`.
4. Recompute h_{j+1} .
5. Repeat for all subsequent entries.

The result is an internally consistent forged chain that passes all single-chain checks.

Remark. In v0.3a, this is documented as a known limit (deliberate negative test `test_cascading_tamper_is_not` in the test suite).

8.4 Multi-party hash graph defence

The architecture’s LEG 7 calls for a hash *graph* (DAG) instead of a chain. Each party maintains its own per-party chain. Entries cross-reference entries in other parties’ chains. Every entry is signed by its producing party.

To forge the graph, an attacker would need to:

1. Modify their own entry.
2. Forge signatures from all parties whose entries reference theirs.
3. Recursively forge signatures up the cross-reference graph.

This requires compromising multiple independent parties’ signing keys simultaneously — much harder than single-key cascading.

Bitcoin’s defence against cascading tamper comes from proof-of-work plus distributed consensus. Our protocol’s defence (in the eventual LEG 7 design) comes from multi-party signatures plus cross-reference structure. Both protect against cascading; they use different cryptographic primitives.

9 End-to-end security chain

The protocol’s security argument composes the pieces:

1. **Circuit family is approximately Haar-random.** At $d = 10$ with all-to-all connectivity, the family is close to a 2-design — PT statistics hold (empirically validated).
2. **F_{XEB} measures fidelity.** Under PT, $F_{\text{XEB}} \approx F$. Empirically: noiseless $F_{\text{XEB}} \approx 1$, noisy $F_{\text{XEB}} \approx F$.
3. **Two-sided XEB check rejects classical attacks.** χ_{low} rejects circuit-blind attacks; χ_{high} rejects heavy-output attacks.

4. **Passing F_{XEB} implies min-entropy.** Under LLHA and Liu et al. 2025 Theorem 5.2, passing $F_{\text{XEB}} \geq \chi$ implies $H_{\min}^{\varepsilon_s} \geq Q_{\min}(n-1) + \log \varepsilon_s$.
5. **Leftover Hash Lemma extracts uniform bits.** Given $H_{\infty} \geq k$ and soundness ε , the Toeplitz extractor (with independent drand seed) produces up to $k - 2\log_2(1/\varepsilon)$ bits ε -close to uniform.
6. **Audit log makes the chain verifiable.** Every cryptographic artifact is recorded with hash-chain integrity. Any party can independently verify each step.

Each step is a separate piece of math with its own assumptions and limits. The composition gives the final claim: *the output bits are unforgeable, unpredictable, and ε -close to uniform under the protocol's stated assumptions.*

10 Open mathematical questions

Items needing cryptographer input before the spec is rigorous:

1. **VRF construction.** Sign+Hash (v0.x) vs. RFC 9381 ECVRF vs. threshold BLS. Affects formal pseudorandomness claims.
2. **AH bound to use.** Honest-server simplification vs. full $Q_{\min}$ computation. At what parameter scales is each appropriate?
3. **LLHA parameter B .** Currently bounded by $B \leq n/2$ (Grover). Tighter bounds for our specific circuit family?
4. **Extractor strength.** Toeplitz is 2-universal (classical). Liu et al. use “quantum-proof strong extractor.” Do we need the stronger primitive given our threat model?
5. **Sample independence.** The $(n-1)$ -per-sample bound assumes independence. Realistic at our parameters, or correlated analysis needed?
6. **Smoothness parameter ε_s .** Liu et al. use $\varepsilon_s = \varepsilon_{\text{sou}}/4$. Tunable for our protocol?
7. **Multi-shot vs. single-shot model.** AH 2023 uses many samples per circuit; Liu et al. uses one. Which model does our protocol commit to?

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